

PHYS 486 - HW05

1.a) $[T_L, x] = T_L x - x T_L$

$$\left. \begin{aligned} T_L (x f(x)) &= (x-L) f(x-L) \\ x T_L f(x) &= x f(x-L) \end{aligned} \right\} \begin{aligned} [T_L, x] f(x) &= (x-L) f(x-L) - x f(x-L) \\ &= -L f(x-L) \\ &= -L T_L f(x) \end{aligned}$$

Thus, $[T_L, x] = -L T_L$

1.b) Let us expand $f(x-L)$ around x . Taylor expansion of $f(y)$ around $y=a$ is

$$f(y) = f(a) + (y-a) f'(a) + \frac{(y-a)^2}{2!} f''(a) + \frac{(y-a)^3}{3!} f'''(a) + \dots$$

They plugging in $y = x-L$, $a = x$

$$f(x-L) = f(x) + (-L) f'(x) + \frac{(-L)^2}{2!} f''(x) + \dots = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!} (-L)^n$$

$$\begin{aligned} T_L f(x) &= \left[1 + (-L) \frac{\partial}{\partial x} + \frac{(-L)^2}{2!} \frac{\partial^2}{\partial x^2} + \dots \right] f(x) \\ &= \sum_{n=0}^{\infty} \frac{(-L \frac{\partial}{\partial x})^n}{n!} f(x) = e^{-L \frac{\partial}{\partial x}} f(x) \end{aligned}$$

Thus, $T_L = e^{-L \frac{\partial}{\partial x}}$ $\square$

$$\hat{p} = -i\hbar \frac{\partial}{\partial x} \Rightarrow -\frac{\partial}{\partial x} = \frac{\hat{p}}{i\hbar} = -\frac{i\hat{p}}{\hbar} \Rightarrow T_L = e^{-iL\hat{p}/\hbar}$$

$$1.c) \quad f(x) = \int_{-\infty}^{\infty} dk \frac{e^{ikx}}{\sqrt{2\pi}} \tilde{f}(k)$$

$$T_L f(x) = f(x-L) = \int_{-\infty}^{\infty} dk \frac{e^{ik(x-L)}}{\sqrt{2\pi}} \tilde{f}(k) \quad \begin{array}{l} * \text{Change of variables} \\ x \rightarrow x-L \end{array}$$

$$= \int_{-\infty}^{\infty} dk \frac{e^{ikx}}{\sqrt{2\pi}} e^{-ikL} \tilde{f}(k) = \int_{-\infty}^{\infty} dk \frac{e^{ikx}}{\sqrt{2\pi}} \tilde{f}_L(k)$$

Fourier Transform of $f(x-L)$

$$\tilde{f}_L(k) = e^{-ikL} \tilde{f}(k)$$

Shift by L in spatial domain is equivalent to multiplying $\tilde{f}(k)$ by e^{-ikL} (a phase factor).

$$1.d) \quad [\hat{p}, T_L] = \hat{p} e^{-iL\hat{p}/\hbar} - e^{-iL\hat{p}/\hbar} \hat{p} = 0$$

$$\hat{p} e^{-iL\hat{p}/\hbar} = e^{-iL\hat{p}/\hbar} = \sum_{n=0}^{\infty} \frac{(-iL/\hbar)^n}{n!} \hat{p}^{n+1}$$

$\hookrightarrow$ Since $e^{-iL\hat{p}/\hbar} = \sum_{n=0}^{\infty} \frac{(-iL/\hbar)^n}{n!} \hat{p}^n$ is just a linear combination of momentum operators, $\hat{p}_L$ commutes with $\hat{T}_L$.

Since $\hat{T}_L$ commutes with $\hat{p}_L$, they share a common eigenbasis. Thus, they share common eigenfunctions.

$\hookrightarrow$ Refer to Discussion 5, Question 4.

* We've showed that shifting the wavefunction in spatial domain only brings a phase factor e^{-ikL} . However, observable probability of momentum doesn't change

$$P(k) = |\tilde{f}_L(k)|^2 = |\tilde{f}(k) e^{-ikL}|^2 = |\tilde{f}(k)|^2$$

Thus, momentum of particle remains constant under spatial translations. Its wavefunction merely accumulates a phase shift.

→ This is also known as Noether's Theorem, For a system that is invariant under spatial translation $[T_L, \hat{p}] = 0$; momentum is conserved

$$2) \quad E_n = n^3 \hbar \omega, \quad n=1, 2, \dots$$

$$2.a) \quad \hat{H} |n\rangle = E_n |n\rangle = n^3 \hbar \omega |n\rangle$$

$$\left. \begin{aligned} \hat{H} |1\rangle &= \hbar \omega |1\rangle \\ \hat{H} |2\rangle &= 8 \hbar \omega |2\rangle \\ \hat{H} |3\rangle &= 27 \hbar \omega |3\rangle \\ &\vdots \end{aligned} \right\}$$

$$\hat{H} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 8 & & & \\ & & 27 & & \\ & & & 64 & \\ 0 & & & & \ddots \end{bmatrix} = \text{diag}([1, 8, 27, 64, \dots])$$

↳ Matrix is infinite size since eigenstates live in the Hilbert space.

$$2.b) \quad Q |n\rangle = 3n^2 |n+2\rangle$$

$$\left. \begin{aligned} Q |1\rangle &= 3 |3\rangle \\ Q |2\rangle &= 12 |4\rangle \\ Q |3\rangle &= 27 |5\rangle \\ Q |4\rangle &= 48 |6\rangle \end{aligned} \right\}$$

$$Q = \begin{bmatrix} & & & & & 0 \\ & & & & & \\ & & & & & \\ 3 & & 0 & 0 & & \\ & 12 & 0 & 0 & & \\ & & 27 & 0 & 0 & \\ & & & 48 & 0 & 0 \\ & & & & \ddots & \\ 0 & & & & & \end{bmatrix}$$

$$Q_{ij} = 3(j)^2 \delta_{i+2,j}$$

Infinite matrix with $3n^2$ sequence, 2 below main diagonal.

2.c)

$$\hat{H} \hat{Q} |n\rangle = \hat{H} 3n^2 |n+2\rangle = 3n^2 (n+2)^3 \hbar \omega |n+2\rangle$$

$$\hat{Q} \hat{H} |n\rangle = \hat{Q} n^3 \hbar \omega |n\rangle = n^3 \hbar \omega 3n^2 |n+2\rangle$$

$$[\hat{H}, \hat{Q}] |n\rangle = 3n^2 \hbar \omega [(n+2)^3 - n^3] |n+2\rangle \neq 0 \Rightarrow \hat{H}, \hat{Q} \text{ don't commute}$$

→ Q is NOT Hermitian since $Q^\dagger = (Q^*)^T = Q^T \neq Q$
 Q is not symmetric due to off-diagonal

$$3) \quad H \Psi(\vec{x}) = -\frac{\hbar^2}{2m} \nabla^2 \Psi(\vec{x}) = E \Psi(\vec{x}) \quad : \text{Free-particle}$$

$$3.a) \quad p_x p_y f = -i\hbar \frac{\partial}{\partial x} \left(-i\hbar \frac{\partial}{\partial y} f \right) = -\hbar^2 \frac{\partial^2 f}{\partial x \partial y} = -\hbar^2 \frac{\partial^2 f}{\partial y \partial x} \\ = p_y p_x f$$

$\hookrightarrow [p_x, p_y] = 0$ since derivative operators for orthogonal directions commute with each other. Furthermore

$$[p_i, p_j] = 0, \quad \forall i, j \in \{x, y, z\} \quad \text{due to symmetry}$$

Writing the gradient op. in Hamiltonian i.t.o. derivatives

$$H \Psi(\vec{x}) = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \Psi(\vec{x}) = \frac{p_x^2 + p_y^2 + p_z^2}{2m} \Psi(\vec{x})$$

$$[p_x, H] = \frac{1}{2m} \left(\underbrace{[p_x, p_x^2]}_0 + \underbrace{[p_x, p_y^2]}_0 + \underbrace{[p_x, p_z^2]}_0 \right) = 0$$

$\rightarrow$ Note that $[p_i, p_j^2] = 0$ since $[p_i, p_j] = 0$

$$\text{Similarly } [p_y, H] = [p_z, H] = [p_x, H] = 0$$

$\hookrightarrow$ All momentum ops. commute with Hamiltonian.

$$3.b) \quad \vec{L} = \vec{x} \times \vec{p} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix} = (y p_z - z p_y) \hat{x} + (z p_x - x p_z) \hat{y} + (x p_y - y p_x) \hat{z}$$

Let us test $L_x = y p_z - z p_y$ component

$$[H, L_x] = \frac{1}{2m} [p_x^2 + p_y^2 + p_z^2, y p_z - z p_y]$$

→ Notice that p_x^2 term commutes with anything containing y, p_y, z, p_z since orthogonal directions are independent (derivatives commute)

$$[H, L_x] = \frac{1}{2m} \left([p_y^2, y p_z] - \underbrace{[p_y^2, z p_y]}_{=0} + \underbrace{[p_z^2, y p_z]}_{=0} - [p_z^2, z p_y] \right)$$

$$z [p_y^2, p_y] = 0 \quad y [p_z^2, p_z] = 0$$

↳ Position term can go out of commutator since $p_y^2 z = z p_y^2 \Rightarrow$ orthogonal component commute

$$[H, L_x] = \frac{1}{2m} \left([p_y^2, y p_z] - [p_z^2, z p_y] \right)$$

$$[p_y^2, y p_z] = [p_y^2, y] p_z \quad \text{and using identity } [\hat{A}\hat{B}, \hat{C}] = [\hat{A}, \hat{C}]\hat{B} + \hat{A}[\hat{B}, \hat{C}]$$

$$\bullet [p_y^2, y] p_z = \underbrace{[p_y, y]}_{-i\hbar} p_y p_z + p_y \underbrace{[p_y, y]}_{-i\hbar} p_z = -2i\hbar p_y p_z$$

$$\bullet [p_z^2, z p_y] = -2i\hbar p_z p_y \quad \text{by symmetry}$$

$$[H, L_x] = \frac{1}{2m} (-2i\hbar) (\underbrace{p_y p_z}_{p_z p_y} - p_z p_y) = 0$$

→ Similarly $[H, L_i] = 0, \forall i \in \{x, y, z\}$ due to symmetry

Since H commutes with $\vec{L}$ it must also commute with L^2 . One simple way to see this is the following

$$[H, L^2] = [H, \vec{L}] \cdot \vec{L} + \vec{L} \cdot [H, \vec{L}] = 0 \quad \text{using commutator identity.}$$

3.c) Recall we've found:

$$\vec{L} = \vec{r} \times \vec{p} = (y p_z - z p_y) \hat{x} + (z p_x - x p_z) \hat{y} + (x p_y - y p_x) \hat{z}$$

$$\vec{L}^\dagger = (p_z^\dagger y^\dagger - p_y^\dagger z^\dagger) \hat{x} + (p_x^\dagger z^\dagger - p_z^\dagger x^\dagger) \hat{y} + (p_y^\dagger x^\dagger - p_x^\dagger y^\dagger) \hat{z}$$

Since position and momentum operators are Hermitian $\begin{cases} \{x, y, z\}^\dagger = \{x, y, z\} \\ \{p_x, p_y, p_z\}^\dagger = \{p_x, p_y, p_z\} \end{cases}$ and position-momentum operators of orthogonal directions commute we have:

$$p_z^\dagger y^\dagger = y p_z, \quad p_y^\dagger z^\dagger = z p_y \quad \text{and similarly for other terms}$$

$$\text{Then, } \vec{L}^\dagger = (y p_z - z p_y) \hat{x} + \dots = \vec{L} \Rightarrow \underline{\vec{L} \text{ is Hermitian}}$$

OR more formally

$$L_i = \epsilon_{ijk} x_j p_k, \quad L_i^\dagger = \epsilon_{ijk} p_k^\dagger x_j^\dagger$$

$$\text{Notice } [x_j, p_k] = x_j p_k - p_k x_j = i\hbar \delta_{jk}$$

$$L_i^\dagger = \epsilon_{ijk} p_k x_j = \epsilon_{ijk} x_j p_k - i\hbar \underbrace{\epsilon_{ijk} \delta_{jk}}_0 = L_i$$

3.d) $\rightarrow$ We have found that $[H, L_z] = [H, L^2] = 0$ in part b

$$\begin{aligned} \rightarrow \text{Let us evaluate } [L_x, L_y] &= [y p_z, z p_x] - [y p_z, x p_z] - [z p_y, z p_x] + [z p_y, x p_z] \\ &= y p_x [p_z, z] - y x \underbrace{[p_z, p_z]}_0 - z^2 \underbrace{[p_y, p_x]}_0 + p_y x [z, p_z] \\ &= i\hbar (x p_y - y p_x) = i\hbar L_z \end{aligned}$$

$$\text{In general } [L_i, L_j] = \epsilon_{ijk} i\hbar L_k$$

$$\rightarrow [L^2, L_z] = [L_x^2, L_z] + [L_y^2, L_z] + \cancel{[L_z^2, L_z]}_0$$

$$= L_x [L_x, L_z] + [L_x, L_z] L_x + L_y [L_y, L_z] + [L_y, L_z] L_y$$

$$= L_x (-i\hbar L_y) + (-i\hbar L_y) L_x + L_y (i\hbar L_x) + (i\hbar L_x) L_y$$

$$[L^2, L_z] = 0,$$

$\rightarrow$ Now we evaluate the commutation relations of parity operator P

$$\left. \begin{aligned} P H \psi(\vec{x}) &= P E \psi(\vec{x}) = E \psi(-\vec{x}) \\ H P \psi(\vec{x}) &= H \psi(-\vec{x}) = E \psi(-\vec{x}) \end{aligned} \right\} [P, H] = 0$$

change of variables
in Schrödinger eq.

$$\begin{aligned} P \vec{L} \psi(\vec{x}) &= P (\vec{x} \times \vec{p}) \psi(\vec{x}) = (P \vec{x} \psi(\vec{x})) \times (P \vec{p} \psi(\vec{x})) \\ &= (-\vec{x} \psi(-\vec{x})) \times \underbrace{\left(-i\hbar \frac{\partial}{\partial x_i} \psi(-\vec{x}) \right)}_{-\vec{p} \psi(-\vec{x})} = (\vec{x} \times \vec{p}) \psi(-\vec{x}) \end{aligned}$$

$$\vec{L} P \psi(\vec{x}) = (\vec{x} \times \vec{p}) \psi(-\vec{x}) \Rightarrow [P, \vec{L}] = 0$$

$$\text{Thus } [P, L_x] = [P, L^2] = 0$$

* Therefore, $\{H, L^2, L_z, P\}$ commutes with each other and they form a set of mutually commuting observables with a shared set of eigenstates.

4. a) $H|n\rangle = (n + 1/2) \hbar \omega$

Let $\hat{X}$ be the matrix representing the position operator.

$$\hat{X}_{nm} = \langle n | \hat{X} | m \rangle, \quad \left. \begin{aligned} a &= \sqrt{\frac{m\omega}{2\hbar}} \hat{X} + i \sqrt{\frac{1}{2m\omega\hbar}} \hat{p} \\ a^\dagger &= \sqrt{\frac{m\omega}{2\hbar}} \hat{X} - i \sqrt{\frac{1}{2m\omega\hbar}} \hat{p} \end{aligned} \right\} \hat{X} = \sqrt{\frac{\hbar}{2m\omega}} (a + a^\dagger)$$

$$\begin{aligned} \hat{X}_{nm} &= \langle n | \sqrt{\frac{\hbar}{2m\omega}} (a + a^\dagger) | m \rangle = \sqrt{\frac{\hbar}{2m\omega}} \left(\langle n | \sqrt{m} | m-1 \rangle + \langle n | \sqrt{m+1} | m+1 \rangle \right) \\ &= \sqrt{\frac{\hbar}{2m\omega}} \left(\sqrt{m} \langle n | m-1 \rangle + \sqrt{m+1} \langle n | m+1 \rangle \right) \end{aligned}$$

Since eigenstates of the simple harmonic oscillator are orthonormal

$$\langle n | m \rangle = \delta_{n,m} \Rightarrow \hat{X}_{nm} = \sqrt{\frac{\hbar}{2m\omega}} \left(\sqrt{m} \delta_{n,m-1} + \sqrt{m+1} \delta_{n,m+1} \right)$$

$$\hat{X}_{12} = \sqrt{\hbar} , \quad \hat{X}_{21} = \sqrt{4\hbar} , \quad \hat{X}_{23} = \sqrt{6\hbar} , \quad \hat{X}_{32} = \sqrt{6\hbar} \dots$$

$$\hat{X} = \sqrt{\frac{\hbar}{2m\omega}} \begin{bmatrix} 0 & \sqrt{2} & & & 0 \\ \sqrt{1} & 0 & \sqrt{3} & & \\ & \sqrt{2} & 0 & \sqrt{4} & \\ & & \sqrt{3} & 0 & \sqrt{5} \\ & & & \sqrt{4} & 0 & \sqrt{6} \\ & & & & \ddots & \ddots \\ 0 & & & & & \ddots \end{bmatrix}$$

$$\hat{X}^\dagger = (\hat{X}^*)^T = \underbrace{\hat{X}^T}_{\text{symmetric}} = \hat{X} \Rightarrow \hat{X} \text{ is Hermitian}$$

4.b) Similarly $\hat{p} = i \sqrt{\frac{m\hbar\omega}{2}} (a^\dagger - a)$

$P_{nm} = \langle n | p | m \rangle$: matrix representation of momentum operator.

$$P_{nm} = i \sqrt{\frac{m\hbar\omega}{2}} \langle n | a^\dagger - a | m \rangle = i \sqrt{\frac{m\hbar\omega}{2}} [\sqrt{m+1} \delta_{n,m+1} - \sqrt{m} \delta_{n,m-1}]$$

$$P_{nm} = i \sqrt{\frac{m\hbar\omega}{2}} \begin{bmatrix} 0 & -\sqrt{1} & & & \\ \sqrt{1} & 0 & -\sqrt{2} & & \\ & \sqrt{2} & 0 & -\sqrt{3} & \\ & & \sqrt{3} & 0 & -\sqrt{4} \\ & & & \sqrt{4} & 0 & -\sqrt{5} \\ & & & & \sqrt{5} & 0 & -\sqrt{6} \\ & & & & & \sqrt{6} & 0 & \ddots \\ & & & & & & & \ddots & \ddots \end{bmatrix}$$

$$P_{nm}^* = -P_{nm} = P_{mn} \Rightarrow P^\dagger = P \Rightarrow \hat{p} \text{ is Hermitian.}$$

↘ antisymmetric

4.c) $H = \frac{1}{2m} p^2 + \frac{1}{2} m \omega^2 x^2$

→ In the matrix picture $\hat{p}^2 \longrightarrow P \cdot P$ (matrix mult.)
 $\hat{x}^2 \longrightarrow X \cdot X$

$$P_{ij}^2 = \sum_k P_{ik} P_{kj} = -\frac{m\hbar\omega}{2} \sum_k (\sqrt{k+1} \delta_{i,k+1} - \sqrt{k} \delta_{i,k-1}) (\sqrt{j+1} \delta_{k,j+1} - \sqrt{j} \delta_{k,j-1})$$

Note that

$$\delta_{i,k+1} \delta_{k,j+1} = \delta_{i-1,j+1}$$

$$\delta_{i,k+1} \delta_{k,j-1} = \delta_{ij}$$

$$\delta_{i,k-1} \delta_{k,j+1} = \delta_{ij}$$

$$\delta_{i,k-1} \delta_{k,j-1} = \delta_{i+1,j-1}$$

$$P_{ij}^2 = \frac{-m\omega\hbar}{2} \left[\sqrt{i(j+1)} \delta_{i-1,j+1} - \left(\sqrt{(ij)} + \sqrt{(i+1)(j+1)} \right) \delta_{ij} + \sqrt{(i+1)j} \delta_{i+1,j-1} \right]$$

$$\begin{aligned} X_{ij}^2 &= \sum_k X_{ik} X_{kj} = \frac{\hbar}{2m\omega} \sum_k \left(\sqrt{k+1} \delta_{i,k+1} + \sqrt{k} \delta_{i,k-1} \right) \left(\sqrt{j+1} \delta_{k,j+1} + \sqrt{j} \delta_{k,j-1} \right) \\ &= \frac{\hbar}{2m\omega} \left[i(j+1) \delta_{i-1,j+1} + \left(\sqrt{ij} + \sqrt{(i+1)(j+1)} \right) \delta_{ij} + \sqrt{(i+1)j} \delta_{i+1,j-1} \right] \end{aligned}$$

$$H_{ij} = \frac{P_{ij}^2}{2m} + \frac{1}{2} m \omega^2 X_{ij} = \frac{\omega\hbar}{4} \times 2 \times \left(\sqrt{ij} + \sqrt{(i+1)(j+1)} \right) \delta_{ij}$$

↳ Note that cross terms with $\delta_{i-1,j+1}$ and $\delta_{i+1,j-1}$ perfectly cancel each other.

↳ Only terms left are δ_{ij} (diagonal terms)

H_{ij} is a diagonal matrix with $H_{nn} = \frac{\hbar\omega}{2} (n+n+1) = \hbar\omega \left(n + \frac{1}{2} \right)$

$$H = \hbar\omega \begin{bmatrix} 1/2 & & & 0 \\ & 3/2 & & \\ & & 5/2 & \\ & & & \ddots \end{bmatrix} \Rightarrow$$

We recover the energy eigenvalues of the SHO as being the diagonal elements since matrices we've used adopted energy eigenbasis.